

Big Data Analytics Lab Viva Questions with Answers

1. What is Big Data?

Ans: Big Data refers to extremely large datasets that cannot be processed by traditional systems.

2. What are the 5 V's of Big Data?

Ans: Volume, Velocity, Variety, Veracity, and Value.

3. What is Hadoop?

Ans: Hadoop is an open-source framework for distributed storage and processing of large datasets.

4. What are the main components of Hadoop?

Ans: HDFS, MapReduce, and YARN.

5. What is HDFS?

Ans: HDFS (Hadoop Distributed File System) stores data across multiple machines.

6. What is a block in HDFS?

Ans: A block is the smallest unit of data storage in HDFS, usually 128 MB.

7. What is NameNode?

Ans: NameNode manages metadata and directory structure in HDFS.

8. What is DataNode?

Ans: DataNode stores actual data blocks and sends heartbeats to the NameNode.

9. What happens if a NameNode fails?

Ans: A secondary or standby NameNode can restore metadata backup.

10. What is replication in HDFS?

Ans: Replication means copying each block to multiple nodes for fault tolerance.

11. What is MapReduce?

Ans: MapReduce is a programming model for parallel data processing on Hadoop.

12. What are the phases of MapReduce?

Ans: Map phase and Reduce phase.

13. What is YARN?

Ans: YARN (Yet Another Resource Negotiator) manages cluster resources and job scheduling.

14. What is the difference between Hadoop 1.x and 2.x?

Ans: Hadoop 2.x introduced YARN for better scalability and resource management.

15. What is Hive?

Ans: Hive is a data warehouse tool that uses SQL-like queries (HiveQL) on Hadoop.

16. What is Pig?

Ans: Pig is a high-level scripting platform that converts Pig Latin scripts into MapReduce jobs.

17. What is HBase?

Ans: HBase is a NoSQL database on HDFS for real-time read/write access to Big Data.

18. What is Sqoop?

Ans: Sqoop is used to transfer data between Hadoop and relational databases.

19. What is Flume?

Ans: Flume is used for collecting and transferring large volumes of streaming data to HDFS.

20. What is Oozie?

Ans: Oozie is a workflow scheduler used to manage Hadoop jobs.

21. What is Apache Spark?

Ans: Spark is a fast, in-memory data processing framework that supports batch and real-time analytics.

22. What is RDD?

Ans: RDD (Resilient Distributed Dataset) is a basic data structure in Spark for distributed computing.

23. What is the difference between Spark and Hadoop MapReduce?

Ans: Spark processes data in memory; MapReduce writes intermediate data to disk.

24. What is the role of Zookeeper?

Ans: Zookeeper coordinates distributed applications and manages configuration information.

25. What is speculative execution in Hadoop?

Ans: It runs duplicate tasks on slower nodes to prevent job delays.

26. What is a Combiner in MapReduce?

Ans: A mini-reducer that performs local aggregation before the reduce phase.

27. What is Partitioner in MapReduce?

Ans: It controls how the intermediate key-value pairs are distributed to reducers.

28. What is a Data Lake?

Ans: A storage system that holds raw data in its native format until needed.

29. What is Data Cleaning?

Ans: The process of correcting or removing inaccurate or incomplete data.

30. What is the function of ResourceManager in YARN?

Ans: It allocates resources and manages the overall cluster workload.

31. What is NodeManager?

Ans: NodeManager runs on each node and manages resources locally.

32. What is schema-on-read?

Ans: Data is stored as-is and schema is applied when reading the data.

33. What is schema-on-write?

Ans: Schema is applied before storing the data.

34. What is a JobTracker?

Ans: In Hadoop 1.x, JobTracker manages and schedules MapReduce jobs.

35. What is a TaskTracker?

Ans: It executes tasks as directed by the JobTracker on each node.

36. What are the three modes of Hadoop?

Ans: Standalone, Pseudo-distributed, and Fully-distributed mode.

37. What is the default replication factor in Hadoop?

Ans: The default replication factor is 3.

38. What is a DataNode heartbeat?

Ans: A periodic signal sent by DataNodes to NameNode to indicate they are active.

39. What command checks Hadoop version?

Ans: Use 'hadoop version' command.

40. What command lists files in HDFS?

Ans: Use 'hdfs dfs -ls /' command.

41. What command copies a file to HDFS?

Ans: Use 'hdfs dfs -put localfile /destinationpath'.

42. What is the command to remove a file in HDFS?

Ans: Use 'hdfs dfs -rm filename'.

43. What is the command to check HDFS disk usage?

Ans: Use 'hdfs dfs -du -h /'.

44. What are the advantages of Hadoop?

Ans: Scalability, fault tolerance, cost-effectiveness, and flexibility.

45. What are the disadvantages of Hadoop?

Ans: High latency, complex setup, and not suitable for small data.

46. What is the role of Secondary NameNode?

Ans: It merges NameNode logs to prevent metadata overflow.

47. What is HiveQL?

Ans: HiveQL is a SQL-like query language used in Hive.

48. What is the difference between Hive and Pig?

Ans: Hive uses SQL-like syntax, while Pig uses a scripting language (Pig Latin).

49. What are the types of joins in Hive?

Ans: INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN.

50. What is checkpointing in Hadoop?

Ans: Checkpointing saves metadata snapshots to recover NameNode faster.